Enhancing STEM Education Accessibility for Blind and Low Vision Students

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***Abstract*—This study focuses on enhancing STEM education accessibility for blind and low vision students through the development of innovative assistive technologies. The primary challenge addressed is the limited engagement and effectiveness of traditional teaching methods for visually impaired students, particularly in science and mathematics. Leveraging advancements in computer vision, machine learning, and internet of things technologies, the proposed solution includes a mobile application integrated with YOLO-based object detection and real-time audio feedback. Methods employed involve capturing and preprocessing images of geometric shapes, educational materials, and currency, followed by rigorous model training using Google Colab. The system's architecture incorporates YOLOv8 for real-time object detection, combined with text-to-speech and speech-to-text technologies to provide comprehensive audio descriptions and interactive feedback. Results demonstrate significant improvements in detection accuracy and user engagement, with YOLOv8 achieving high precision in object recognition tasks. Performance standards were evaluated through confusion matrices, highlighting the model's robustness across various educational scenarios. The developed system effectively enhances independent learning and engagement among visually impaired students, promoting inclusivity and equity in STEM education.**

***Keywords*—assistive technology, STEM education, object detection, visually impaired, low vision, blind, YOLOv8, Google Colab**

I. INTRODUCTION

Eyes are essential for most human activities, including learning, but some people are unfortunately deprived of this ability. According to the World Health Organization, approximately 284 million people worldwide have visual impairments, with 39 million of them being completely blind [1]. In Sri Lanka, around 1.7% of the population, which equates to about 364,000 individuals, are visually impaired [2]. A survey highlighted that 1.6% of this group have severe visual impairments, while 15.4% have some degree of visual impairment [3]. The Ceylon School for the Deaf and Blind, established in 1912, marked the commencement of educational efforts for the blind in Sri Lanka [2]. Despite a commendable literacy rate of 92.5%, the country still struggles with Braille literacy. According to the Ministry of Social Welfare, 71% of blind or visually impaired individuals have attended school to some extent, yet only 6% have progressed beyond the General Certificate of Education (G.C.E.) Ordinary Level [4].

Enhancing the education system to better serve visually impaired students is crucial to making learning more accessible and engaging, thereby increasing their interest in education. This study addresses the problem of developing assistive technologies to improve the educational experiences of visually impaired students, particularly in subjects like science, mathematics, and object recognition. Advances in computer vision, machine learning, and accessibility technologies can be leveraged to create innovative solutions that enhance accessibility for blind individuals.

Currently, visually impaired students rely on tools such as embossed images, real objects, and Braille books. Traditional teaching methods can be time-consuming and often fail to capture students' interest, leading to low engagement levels. Adapting to new technologies can be challenging, but doing so can potentially boost students' interest in learning. Compared to their sighted peers, visually impaired students often find learning more difficult. Teachers from schools for the deaf and blind have noted that subjects like science and mathematics pose additional challenges. Enhancing memory retention and maintaining focus can be more effectively achieved through engaging activities rather than traditional methods. However, it's important to acknowledge the potential risks associated with tactile learning of geometric and scientific objects, which may cause injury. Additionally, the discomfort visually impaired students experience with money handling underscores the need for early financial literacy education. Such programs can provide essential skills and confidence, facilitating smoother financial transactions.

Recognizing the role of technology in education, a smart educational application combined with an Internet of Things (IoT) device proves to be a valuable tool for supporting the learning needs of visually impaired students. Object detection, paired with a user-friendly mobile application, offers an effective solution to enhance learning activities and outcomes for these students. Furthermore, as technology continues to advance, its role in education will only grow. It is imperative that the younger generation not only learns to consume this technology but also understands and controls it, ensuring they are not left behind in an increasingly digital world [5].

Section 2 details the background and the literature review, highlighting the current state and limitations of educational aids for visually impaired students. In Section 3, the methodologies employed to integrate and test these technologies are discussed, providing insights into the

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approaches used for system development and implementation. Section 4 presents the results, followed by Section 5, where a thorough discussion interprets the implications of these findings. Finally, the 'Conclusion and Future Work' section reflects on the impact of these technologies on educational accessibility and outlines potential directions for further research to continue advancing this critical field.

II. LITERATURE REVIEW

Research on educational systems for blind and visually impaired students has been a focus of scholarly interest for many years. Historically, the ancient Egyptians were among the first to address the educational needs of visually impaired learners [1]. The development of assistive technologies has significantly advanced the understanding of object detection for blind students. Early efforts relied on tactile and auditory modalities, such as raised-line drawings and tactile graphics, to help visually impaired individuals understand shapes and diagrams. Over time, these aids evolved to include 3D models and embossed pictures, offering more tangible representations of objects [3], [13].

Despite advancements, only 6% of blind students progress beyond the General Certificate of Education (G.C.E.) Ordinary Level, with science and mathematics posing significant challenges. These students often excel in hands-on activities over traditional note-taking.[12].

The integration of Convolutional Neural Networks (CNNs) has improved the accuracy and efficiency of object detection systems for the visually impaired. The advent of the You Only Look Once (YOLO) real-time object detection system has further enhanced these capabilities, providing rapid and precise object recognition that is particularly suited for real-time applications.

Recent studies highlight the challenges of implementing artificial intelligence (AI) in education, particularly for multilingual students, and emphasize the need for balanced AI-teacher evaluation systems [7]. Additionally, systemic barriers faced by first-generation university students point to the necessity of supportive infrastructures to develop inclusive educational tools [8].

One major issue is the lack of infrastructure and financial resources in schools, which hinders the effective implementation of technology. This problem is compounded by inadequate training for teachers and staff, limiting the integration of technology into the curriculum and restricting visually impaired students' access to essential information [4]. Furthermore, many findings in assistive technology research lack generalizability, contain methodological flaws, and exhibit biases, which challenge their credibility.

Interactive math study systems for blind students are innovative but often limited to currency recognition, which restricts educational access. For example, the Sri Lankan Currency Recognition Device for Visually Impaired People [6] shows progress in currency identification but lacks real-time feedback systems, mobile applications, and interaction with IoT devices, diminishing its effectiveness. Similarly, mobile phone applications [2], for money recognition provide some solutions but often overlook other critical aspects of mathematics instruction [9].

Dunai et al.'s comprehensive study introduced a system capable of recognizing Euro banknotes and providing vocal feedback, significantly enhancing the ability of blind individuals to identify currency [10]. However, this study did not leverage an IoT-based device and could not identify multiple banknotes simultaneously. Its potential impact on mathematics education was limited due to the absence of robust integration with real-world arithmetic problems [12].

Despite improvements, traditional tactile aids such as raised-line drawings and physical models often fail to fully represent complex geometric designs. Digital solutions like screen readers and text-to-speech (TTS) software also struggle with accurately interpreting mathematical notation, making it difficult for visually impaired students to engage with these concepts.

In summary, while significant strides have been made in the field of assistive technology for visually impaired students, many challenges remain. Continued research and collaboration among educators, researchers, and technology developers are essential to ensure equitable access to education for blind students.

III. METHODOLOGY

A. Data Collection Methods

Images of geometric forms commonly used to teach mathematics to blind or visually impaired students were included in the dataset for this study. These images were captured at the Ratmalana Blind School, specifically in classrooms housing fifth and sixth-grade students. Additionally, photographs of various geometric forms, tactile aids, educational materials, science laboratory equipment, and coins and currency notes used during instruction were taken. High-quality digital cameras ensured the images were suitable for subsequent analysis and annotation. Photos of coins and currency notes were also gathered from daily transactions and trips to nearby banks, ensuring a wide variety of denominations and variants commonly encountered in everyday life.

B. Tools and Procedures for Dataset Creation

After data collection, several preprocessing and augmentation techniques were applied to enhance the dataset's quality and diversity. *LabelImg* software facilitated the meticulous annotation process, accurately classifying the geometric shapes visible in the images. This precise annotation enabled the accurate identification and naming of every geometric item in the dataset. The dataset underwent preprocessing using *Roboflow*, a powerful platform designed for data preprocessing and optimization. Techniques such as scaling, grayscale conversion, object isolation, and static cropping were employed to improve image resolution and standardize formats. Augmentation techniques like flipping, rotating, blurring, and adding random noise were used to simulate real-world variations, enhancing the model’s generalization and performance.

C. Theoretical Framework

Three prominent object identification models were considered for the development of the proposed system: Mask R-CNN, YOLO, and Faster R-CNN. Each model has unique features and is suited to different applications based on its strengths and weaknesses. YOLO, known for its real-

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time object detection capabilities, divides the input image into a grid of cells and predicts bounding boxes and class probabilities for each cell simultaneously. Despite sacrificing some accuracy for speed, YOLO is preferred for applications requiring fast processing. Faster R-CNN balances speed and accuracy with a two-stage detection technique, excelling in tasks where accuracy is more critical than real-time speed. Mask R-CNN extends Faster R-CNN's capabilities by adding a segmentation mask prediction branch, offering pixel-level precision, ideal for tasks requiring precise instance segmentation.

D. Development Environments

Google Colab and Jupyter Notebook were used for the development of the proposed system. Google Colab is a cloud-based platform providing free access to GPU and TPU resources, eliminating the need for powerful local hardware and enabling advanced deep learning research. Its collaboration features allow multiple users to work simultaneously on the same notebook, and its integration with Google Drive simplifies dataset access. Jupyter Notebook, an open-source web application, allows users to create and share documents containing live code, equations, visualizations, and narrative text. Running locally, it provides offline access and greater control over computing resources, making it ideal for users valuing customization and flexibility.

E. Hardware and Software Utilized

Raspberry Pi and Arduino boards were employed for different functions due to their unique capabilities. Raspberry Pi, a single-board computer, is ideal for applications requiring substantial computing power, such as media centers and IoT applications, with its robust processor and variety of networking options. Arduino's microcontroller-based design, offering simplicity and real-time control, is excellent for embedded systems and electronics projects involving sensors and actuators. Although Raspberry Pi presents a steeper learning curve, it offers greater flexibility and a broader range of capabilities at a higher cost.

F. Implementation

A mobile application was developed using Flutter for the client-side interface, while the server-side logic was implemented using Python. WebSocket technology enabled real-time, bidirectional, full-duplex communication over a single TCP connection between the Flutter client and the Python server. Flutter facilitated cross-platform mobile app development, ensuring compatibility and a consistent user experience on both iOS and Android devices with a single codebase. Python's versatility, robustness, and comprehensive library support made it the preferred choice for the server-side implementation. WebSocket communication was efficiently implemented using Python's *asyncio* package, allowing the server to manage multiple client connections concurrently while maintaining responsiveness.

G. Proposed System

In the proposed system, all functions utilized the YOLO algorithm implemented using the *OpenCV* library in Python. OpenCV provides robust computer vision capabilities, enabling efficient object detection tasks. *NumPy* was also utilized for mathematical calculations and array manipulation, simplifying the pre-processing and post-processing stages of the object recognition pipeline. The geometric shape detection capabilities were seamlessly integrated with accessibility features to support visually challenged students. The system uses the Flutter TTS library to provide audio feedback, enabling blind users to hear descriptions of recognized geometric shapes in real-time. Additionally, dynamically generated descriptive text for detected shapes is produced by OpenAI GPT-3.5.

The science equipment identification function enhances accessibility by using Flutter TTS for real-time audio descriptions and OpenAI GPT-3.5 for dynamic, relevant content. Blind students can guess equipment names via the Flutter *Speech-to-Text* (STT) library. Gen AI generates random questions for currency detection, with TTS and STT facilitating interaction. This system provides engaging, instructional content, enriching educational experiences and understanding for visually impaired students.

H. User Interaction Flow

As depicted in Fig. 1, the user interaction flow of the system encompasses three key functionalities: Math Object Detection Model (MODM), Laboratory Equipment Detection Model (LEDM), and Currency Recognition Model (CRM), each designed to cater to the unique educational needs of blind and visually impaired students. Students can quickly navigate between these elements from the main menu upon launching the mobile application.

In MODM, students select geometric objects to examine and capture images with camera-equipped smart glasses. The system uses the YOLO object detection paradigm for real-time detections and provides auditory feedback including the name, characteristics, and relevant mathematical equations for calculating parameters such as area, perimeter, and volume. This interactive approach promotes independent learning and comprehension of geometric concepts.

Similarly, students can access LEDM from the application's home page. Once there, users can guess the name of the object by interacting with its interfaces and engaging with science laboratory equipment. Upon accurately guessing an object's name, the system emits an alert stating, "You are correct." If the guessed name is incorrect, the system provides audio feedback, "You are wrong," followed by the correct name and a random description of the particular object.

The CRM enables students to solve mathematical exchange problems from real-world scenarios. They calculate answers and visually demonstrate their responses using physical currency. Employing YOLO object detection, the system verifies the accuracy of calculations and provides real-time audio feedback. If the displayed answer is correct, the system displays "Your answer is correct"; if not, it specifies what the mistake is.

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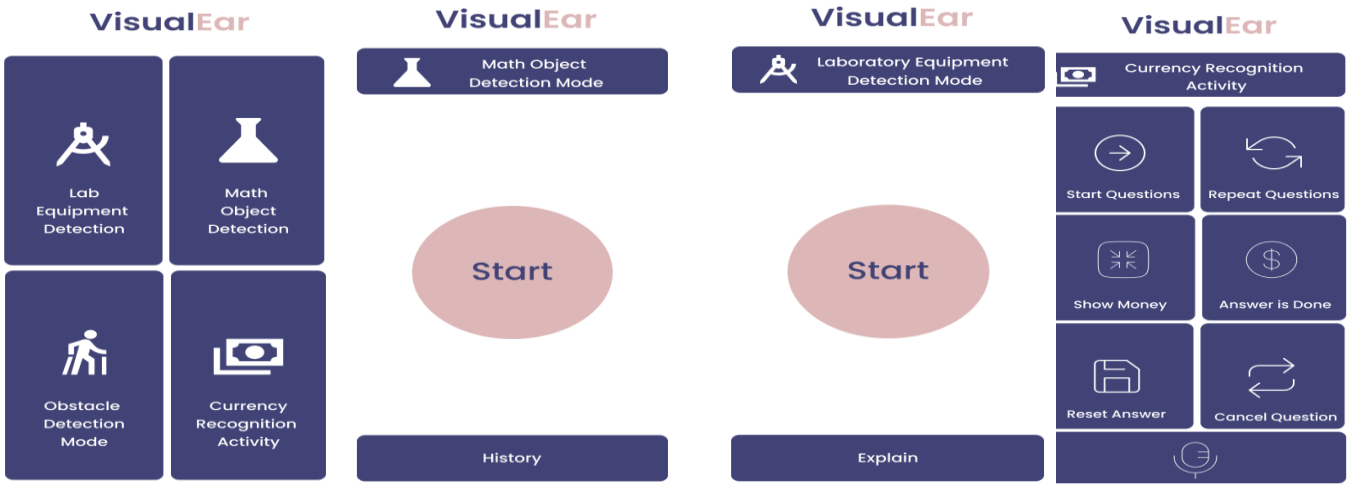


Fig. 1. User interfaces of the proposed system

IV. Results

After conducting a comparative analysis of various YOLO iterations, YOLOv8 emerged as the most effective due to its heightened precision and enhanced training speed. It was therefore chosen to develop three specialized object detection models: the CRM, LEDM, and MODM. This strategy established a uniform architectural basis that improved training efficiency and maximized each model's performance for its specific detection task.

To bolster the models' adaptability and enhance detection accuracy across different environmental conditions and orientations, various data augmentation techniques were applied. Techniques such as image flipping both horizontally and vertically, random image rotations, and color jittering to modify brightness, contrast, and saturation were utilized. The *Roboflow* platform, known for its intuitive data manipulation capabilities, was instrumental in applying these adjustments to the dataset effectively.

A pivotal element of the training methodology involved segregating the dataset into three parts: 70% for training, 20% for testing, and 10% for validation. This separation facilitated continuous monitoring of the learning progress and helped prevent overfitting.

The training sessions were executed on Google Colab, a cloud-based platform that offers an easily accessible environment equipped with Tensor Processing Units (TPUs). TPUs, which are optimized for deep learning tasks, significantly diminished the time required for training and yielded quicker outcomes.

An examination of critical hyperparameters—batch size, image size, and epochs—was undertaken to find the ideal balance that would enhance training effectiveness without compromising model performance. The batch size, which indicates the number of images processed in one iteration, along with the image resolution input into the model, and the epochs, or the total number of complete passes through the training dataset, were adjusted. Experimentation with image sizes ranging from 480x480 to 1280x720, and batch sizes from 1 to 32 over 10 to 50 epochs, led to the selection of a setup with a 16 image batch size, a resolution of 640x480, and 30 epochs. This setup was optimal, efficiently using computational resources and achieving high levels of feature extraction for precise object detection. Fig. 2, 3, and 4 visually represent the experimentation process and the optimal configuration.

As shown in Table 1, the performance of the object detection models was rigorously evaluated using confusion matrices, which detailed the classification accuracy by showing true positives, true negatives, false positives, and false negatives for each model. Fig. 5 and 6 present the confusion matrices for the CRM, LEDM, and MODM. By analyzing these matrices, valuable insights into the models' ability to correctly identify various objects within their respective domains can be gleaned. This information can guide further optimization efforts to enhance the models' performance in real-world applications.

TABLE I. Testing Accuracies for Each YOLO Version

YOLO version	LEDM		CRM		MODM	
	Object detection	Instance segmentation	Object detection	Instance segmentation	Object detection	Instance segmentation
V5	61.32	-	72.00	-	70.99	-
V7	77.35	-	78.97	-	75.43	-
V8	88.67	76.00	83.57	80.32	85.78	81.33



Fig. 2. The training and validation accuracy vs epochs chart of MODM

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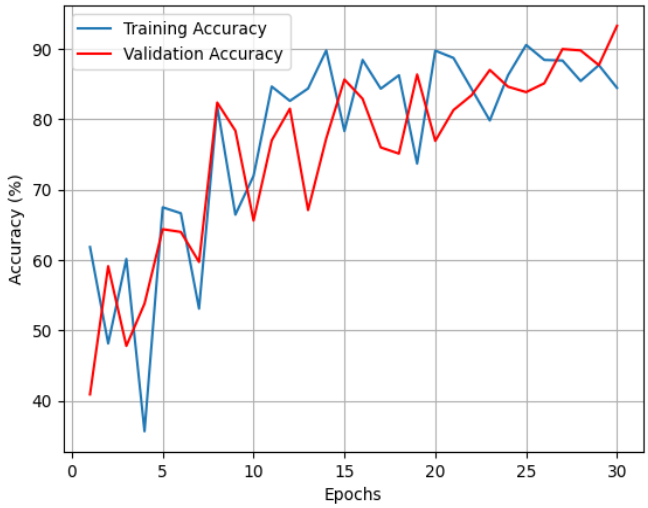


Fig. 3. The training and validation accuracy vs epochs chart of LEDM

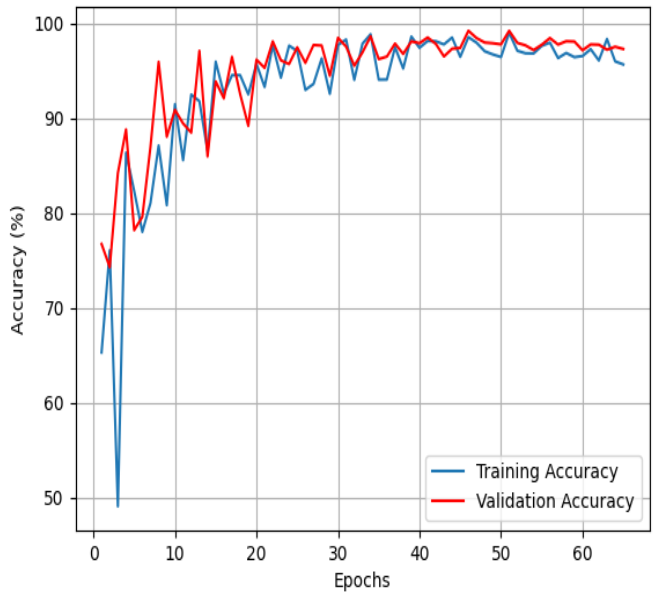


Fig. 4. The training and validation accuracy vs epochs chart of Sri Lankan currency detection model

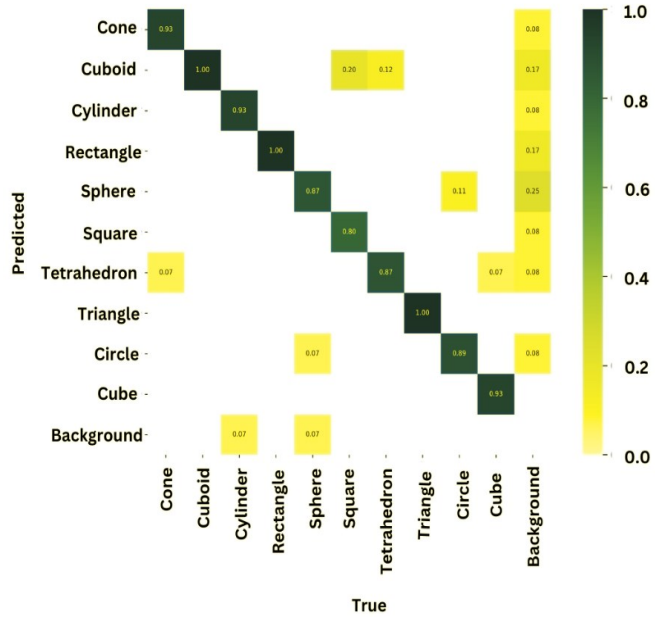


Fig. 5. Confusion matrix normalized chart of math object detection model

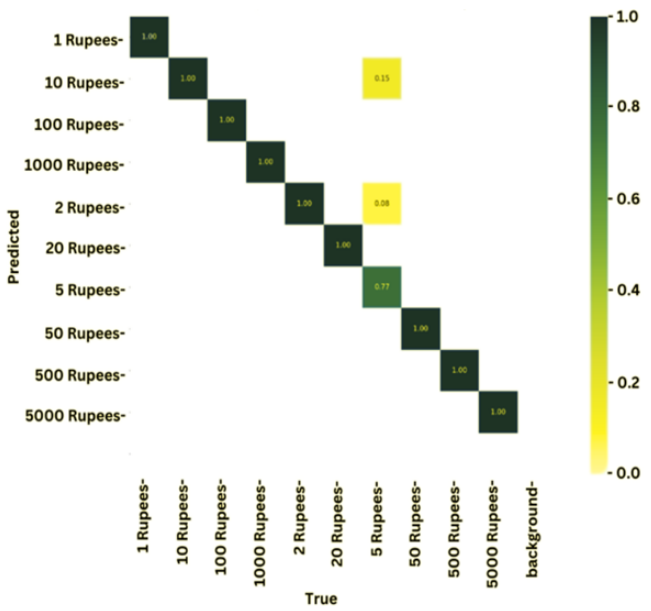


Fig. 6. Confusion matrix normalized chart of Sri Lankan currency detection model

V. Discussion

This research develops an advanced system to revolutionize education for the blind in science, mathematics, and currency usage. Using YOLOv8, STT, and TTS technologies, it empowers blind students to engage in independent learning through accurate object detection and comprehensive descriptions.

A. Addressing Research Gaps

Despite significant strides in assistive technology for visually impaired students, gaps remain. Traditional tactile and auditory aids lack the dynamic interaction of modern computer vision and machine learning. Methods like raised-line drawings and physical models fail to represent complex designs accurately or provide immediate feedback. Existing digital solutions also struggle with mathematical content, as screen readers and text-to-speech software often misinterpret notation, hindering engagement. Moreover, traditional learning processes are time-consuming, leading to slower academic progress and increased isolation.

B. Innovative Contributions

This study introduces several innovative aspects that address the above mentioned gaps:

- Real-Time Object Detection: By employing YOLOv8, the system provides fast and accurate real-time object detection, enhancing the learning experience for visually impaired students by offering immediate feedback and interaction.
- Integrated Educational Tools: The development of a smart educational application combined with IoT devices offers a seamless and interactive learning environment. This integration facilitates a more engaging and efficient learning process compared to traditional methods.
- Comprehensive Audio Feedback: Utilizing TTS technologies, the system delivers detailed auditory descriptions of geometric shapes, scientific equipment, and currency, enabling blind students to receive contextual and comprehensive information in real-time.

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▪ Interactive Learning Models: MODM, Scientific Equipment Detection Model (LEDM), and CRM provide specialized functionalities tailored to the unique educational needs of visually impaired students. These models not only assist in learning but also encourage independent exploration and problem-solving.

▪ Enhanced Accessibility Features: The system's ability to dynamically generate descriptive text using AI technologies like OpenAI GPT-3.5 adds a layer of contextual relevance and engagement that was previously lacking in educational tools for visually impaired students.

By addressing the limitations of past approaches and introducing novel features, this research significantly contributes to assistive technology in education. The findings are crucial for promoting inclusive education, emphasizing equal learning opportunities for all students, regardless of visual abilities. The study aims to enhance inclusivity and equity, removing educational barriers and fostering independence among blind students.

VI. Conclusion and Future Work

This research demonstrates the potential of advanced technologies to transform STEM education for blind and low vision students. The system, using YOLOv8, STT, and TTS, offers real-time object detection and audio descriptions, promoting independent learning and engagement. Despite progress, challenges like infrastructure limitations and teacher training remain.

Future work will focus on developing more robust and scalable solutions inspired by successful models like iPAT [14] to ensure broader accessibility and inclusivity in STEM education. Additionally, integrating language translation technologies such as those discussed in sources [15] and [16] could further enhance educational accessibility for multilingual students. These translation tools can bridge language barriers, ensuring that students from diverse linguistic backgrounds can benefit from the educational resources provided by the system. This integration would be particularly beneficial in multilingual regions, promoting a more inclusive and equitable learning environment for all students. Furthermore, integrating mobile and IoT technologies as explored in [17] can offer innovative ways to enhance real-time interactions and personalized educational experiences for visually impaired students, ensuring they receive timely and context-aware educational support. Moreover, employing machine learning techniques similar to those used in [18] can provide deeper insights into student learning patterns and preferences, enabling the creation of more adaptive and personalized educational tools to better cater to the needs of blind and low vision students.

By addressing these challenges and continuing to innovate, future research can further advance the field of assistive technology in education, ensuring that all students, regardless of their visual or linguistic abilities, have equal access to high-quality STEM education.

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